

Fundamentals of Reinforcement Learning

(1)

→ K-armed Bandit Problems:

✗ Action-values → The value is the expected reward

$$q_*(a) \doteq E[R_t | A_t = a], \forall a \in \{1, \dots, K\}$$

$$q_*(a) = \sum_{\mathcal{X}} p(\mathcal{X}|a) \cdot \mathcal{X}$$

The goal here is to maximise the expected reward.

ie $\boxed{\operatorname{argmax}_a q_*(a)}$

→ Learning action Values:

✗ Estimate the action-values using the sample averaged method.

✗ $q_*(a)$ is not known to the agent. We have to estimate it!

✗ Sample-Averaged Method →

$$Q_t(a) \doteq \frac{\text{Sum of rewards when } a \text{ taken prior to } t}{\text{number of times } a \text{ taken prior to } t}$$

$$= \frac{\sum_{i=1}^{t-1} R_i}{t-1}$$

✓ Greedy Action $\rightarrow$ Selecting action with highest $Q_t(a)$

i.e.
$$a_g = \underset{a}{\operatorname{argmax}} Q(a)$$

✓ Incremental update rule $\rightarrow$

$$Q_{n+1} = \frac{1}{n} \sum_{i=1}^n R_i$$

$$Q_{n+1} = \frac{1}{n} \left(R_n + \sum_{i=1}^{n-1} R_i \right)$$

$$Q_{n+1} = \frac{1}{n} \left(R_n + (n-1) \frac{1}{n-1} \sum_{i=1}^{n-1} R_i \right)$$

$$Q_{n+1} = \frac{1}{n} \left(R_n + Q_n \times (n-1) \right)$$

$$Q_{n+1} = \frac{1}{n} \left(R_n + nQ_n - Q_n \right)$$

$$Q_{n+1} = Q_n + \frac{1}{n} (R_n - Q_n)$$

i.e.
$$\left\{ \begin{array}{l} \text{New estimate} = \text{old estimate} + \\ \text{Step size} \times (\text{target} - \text{old estimate}) \end{array} \right\}$$

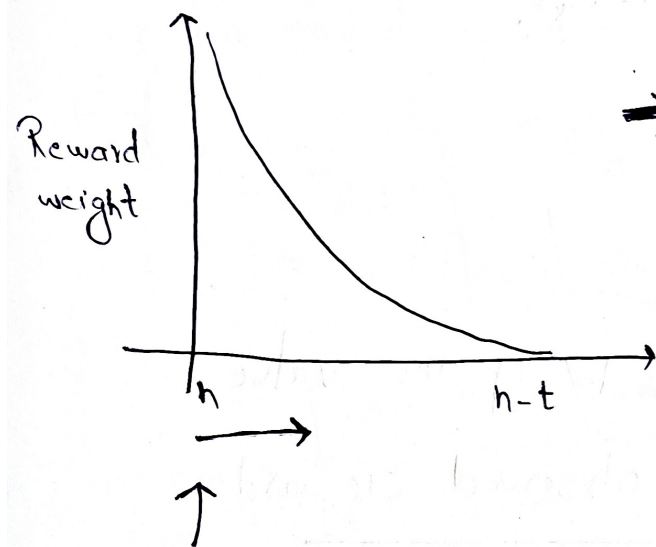
In general —

$$Q_{n+1} = Q_n + \alpha_n (R_n - Q_n)$$

$$\alpha_n \rightarrow [0, 1]$$

✗ In the specific case of Sample average

$$\boxed{\alpha_n = 1/n}$$



Since $\alpha_n \leq 1$,

therefore in case $\alpha_n < 1$,
most recent award
will affect the estimate
more than older rewards.

{ Reward weight for
the Reward received
at time n

Thus the weighting fades
exponentially with time.

$$\Rightarrow Q_{n+1} = Q_n + \alpha_n (R_n - Q_n)$$

$$Q_{n+1} = \alpha R_n + Q_n - \alpha Q_n$$

$$Q_{n+1} = \alpha R_n + (1 - \alpha) Q_n$$

— { α_n is Constant
in time = α }

$$Q_{n+1} = \alpha R_n + (1 - \alpha) [\alpha R_{n-1} + (1 - \alpha) Q_{n-1}]$$

$$Q_{n+1} = \alpha R_n + (1-\alpha)\alpha R_{n-1} + (1-\alpha)^2 Q_{n-1}$$

$\Rightarrow$ performing the same iterative steps we have —

$$Q_{n+1} = \alpha R_n + (1-\alpha)\alpha R_{n-1} + (1-\alpha)^2 \alpha R_{n-2} + \dots + (1-\alpha)^{n-1} \alpha R_1 + (1-\alpha)^n Q_1$$

$$Q_{n+1} = (1-\alpha)^n Q_1 + \sum_{i=1}^n \alpha (1-\alpha)^{n-i} R_i$$

initial action
value

This eqⁿ relates our current estimate of the value Q_{n+1} to Q_1 and all the observed rewards

∅ Exploration and Exploitation trade off $\rightarrow$

1) Exploration — improve knowledge for long term

benefit. By improving the accuracy of the estimated action values, the agent can make more informed decisions in the future.

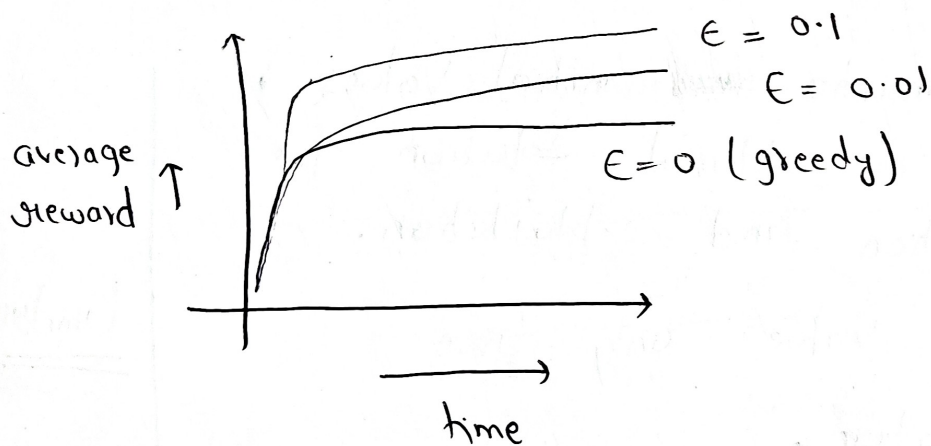
(3)

2) Exploitation - exploit knowledge for short term benefit. It chooses the greedy action to try to get the most reward.

{ However, by being greedy with respect to estimated values, may not actually get the most reward. }

Epsilon - Greedy Action Selection $\rightarrow$

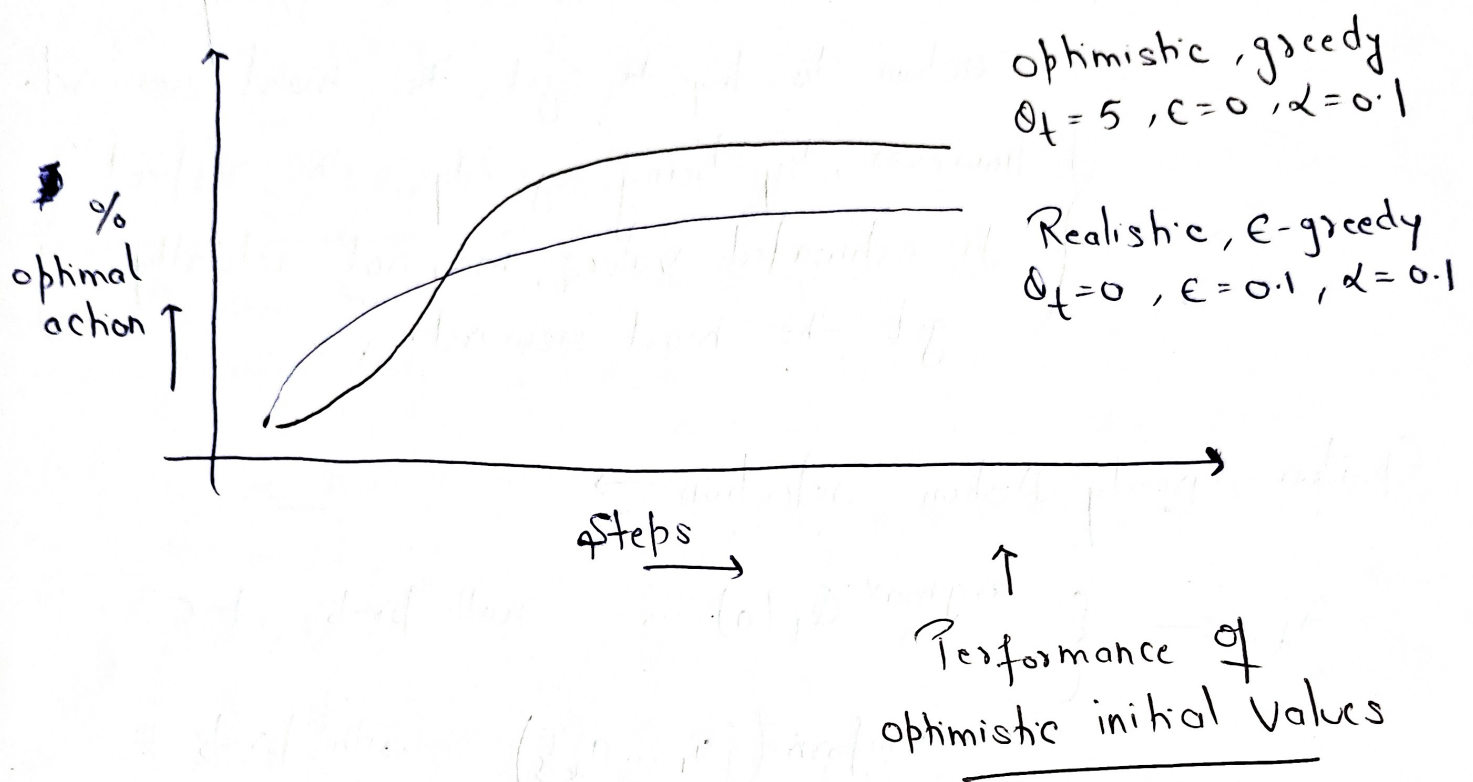
$$A_t \leftarrow \begin{cases} \arg\max_a Q_t(a) & \text{with prob. } 1-\epsilon \\ a \sim \text{Uniform}(\{a_1, \dots, a_k\}) & \text{with prob. } \epsilon \end{cases}$$



Optimistic Initial Values $\rightarrow$

- ✗ How optimistic initial values encourage early exploration.
- ✗ Describe limitations of optimistic initial values

Using optimistic initial values encouraged exploration early in learning.



⇒ However, using optimistic ~~values~~ initial values is not necessarily the optimal solution for balancing exploration and exploitation.

⇒ optimistic initial values only drive early exploration.

⇒ Thus, these are not suited for non-stationary problems.

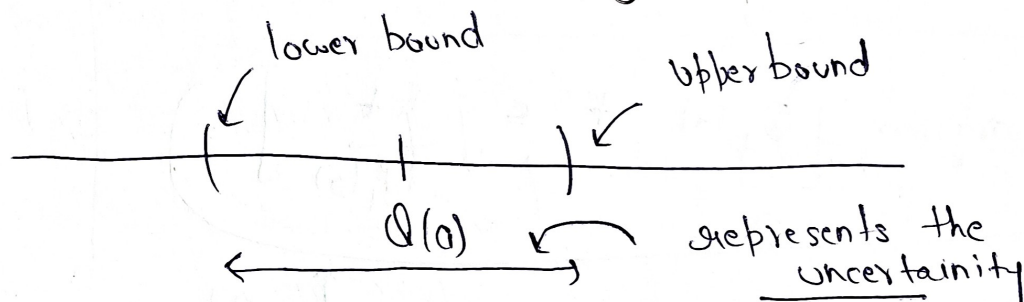
⇒ Also, Sometime we even may not know what the optimistic initial value should be. (because we may not know the maximum reward)

Limitations

(4)
⇒ Regardless of above limitations, optimistic initial values has proven to be very useful heuristic.

Upper Confidence bound Selection →

- ⇒ Another method to balance b/w exploration and exploitation. (like ϵ -greedy)
- ⇒ If we have notation of uncertainty in our value estimates, we can potentially select actions in more intelligent way.
- ⇒ what does it mean to have uncertainty in the estimates? }



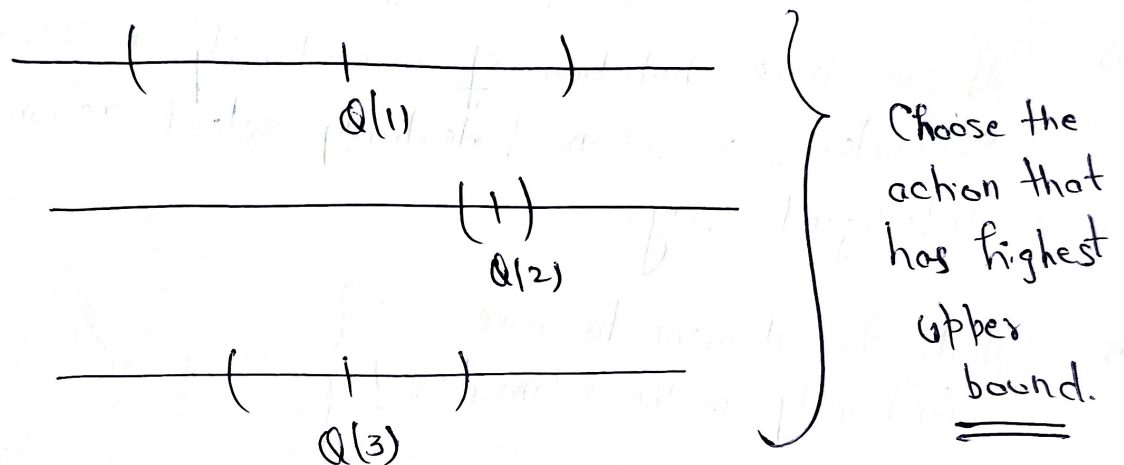
- ✗ $Q(a)$ here represents the current estimate of action 'a'.
- ✗ These brackets represents the confidence interval $Q_{\pm}(a)$. i.e the value of action 'a' lies somewhere in the region.

⇒ If the region is very small ⇒ we are very certain that the value of action 'a' is near our estimated values.

and vice versa

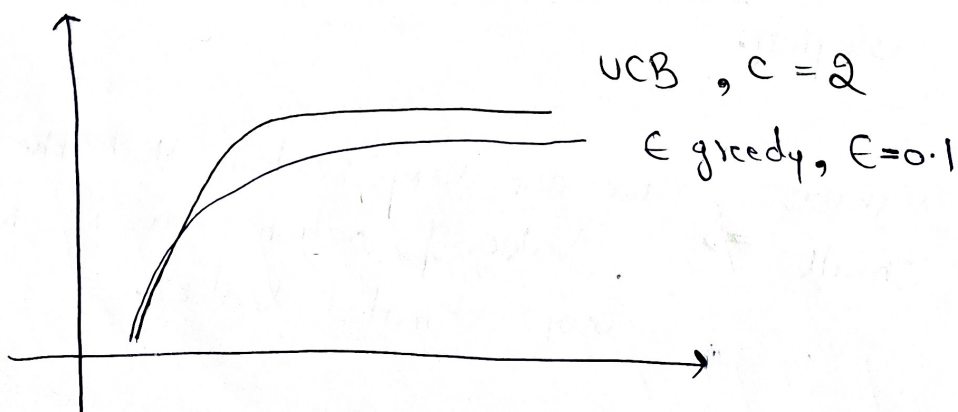
⇒ UCB follows the principle of Optimism in the face of Uncertainty.

i.e. this means if we are uncertain about something than we should optimistically assume that this is good.



$$A_t = \underset{a}{\operatorname{argmax}} \left[Q_t(a) + c \sqrt{\frac{\ln t}{N_t(a)}} \right]$$

↑ exploit
 ↑ controls the amount of exploration.
 ↑ explore



→ Initially, UCB explores more to systematically
reduce uncertainty. UCB exploration
reduces over time whereas ϵ -greedy continues to
take a random action ϵ times. (5)